NAC TO THE FUTURE

Bypassing Network Access Controls and Improving Red Team Implants

AGENDA

01 INTRO + DEFINITIONS

02 BYPASSING NAC

03 ENHANCING EXISTING TOOLS

04 DEFENSE

WHOAMI

Rick Console

- OSCP | CRTO | SecurityX
- Penetration Tester @ Security Risk Advisors
- Develops custom offensive tools + hardware
- Focus: Intersection between AI + Security

GitHub: RickConsole

INTRODUCTION 01

INTRODUCTION

WHY IM HERE

- We frequently encounter 802.1x NAC
- We needed a tool to bypass it

WHAT YOU'LL LEARN

- How 802.1x NAC works
- How to bypass it
- Why so many networks are still vulnerable

ESSENTIAL CONCEPTS

NETWORK ACCESS CONTROL (NAC)

A security system that decides which devices are allowed to connect to a wired or wireless network

NAC BYPASS

Any technique that allows an unapproved device to connect to a network “secured” with NAC

802.1x

A port-based authentication protocol that verifies a device’s identity before allowing it onto the network

MAC ADDRESS

12-digit hexadecimal identifier assigned to a network interface controller (NIC) for communication within a network segment

TYPES OF NETWORK ACCESS CONTROL

NAC	DESCRIPTION	DEPLOYMENT	SECURITY
MAC BASED	Access is granted based solely on a device's MAC address	EASY	WEAK
PORT BASED	The device must authenticate to some auth server to be granted access	MODERATE	MODERATE
AGENT BASED	Agent software running on the device that validates and facilitates the authenticated connection	DIFFICULT	STRONG

BYPASSING NAC 02

MAC-BASED NAC

How It Works

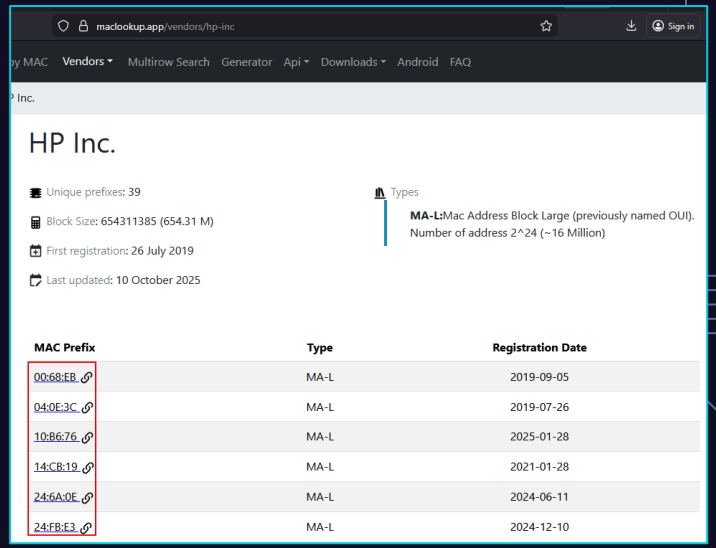
- Security system keeps a list of allow-listed MACs
- That’s it.

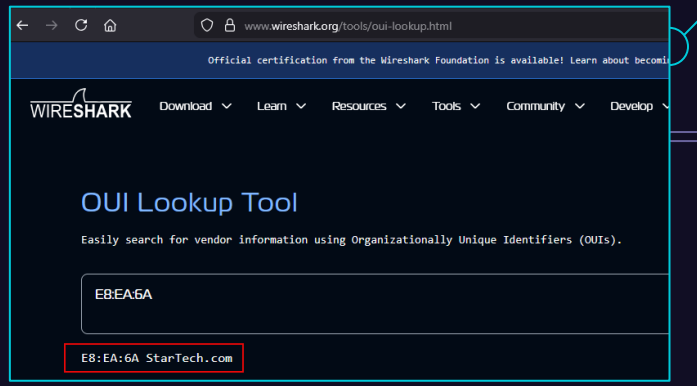
BYPASSING MAC-BASED NAC

- Just LIE
- Listen on the wire for MACs
- Spoof your MAC/OUI (macchanger)
- DONE!

OUIs

- Organizationally Unique Identifier
- Blanket allow-lists
- OUIs can be easily found online
- Helpful for situational awareness





PORT-BASED NAC (802.1x)

How It Works

- Devices must authenticate to a server (RADIUS) in order to be granted network access on the switch
- User/pass or certificate

BYPASSING PORT-BASED NAC (802.1x)

- We place an attacker-controlled device between the supplicant and the switch.
- The switch authenticates the real device as usual.
- Once the port enters “authorized” state, anything sent from that link is trusted.

BYPASSING PORT-BASED NAC (802.1x)

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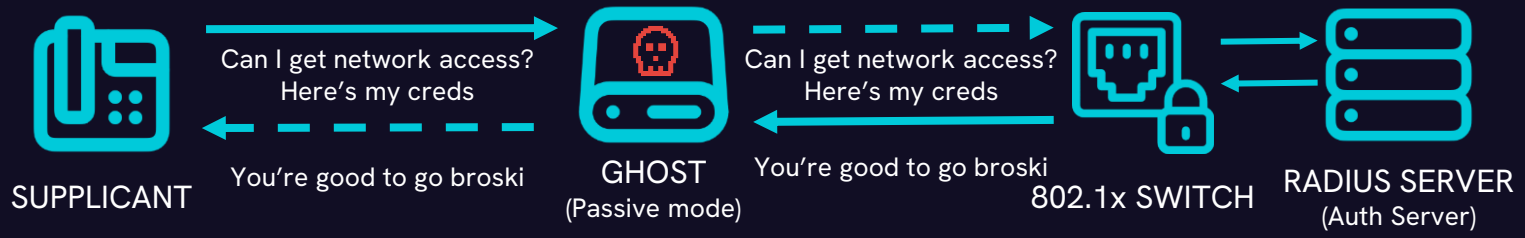

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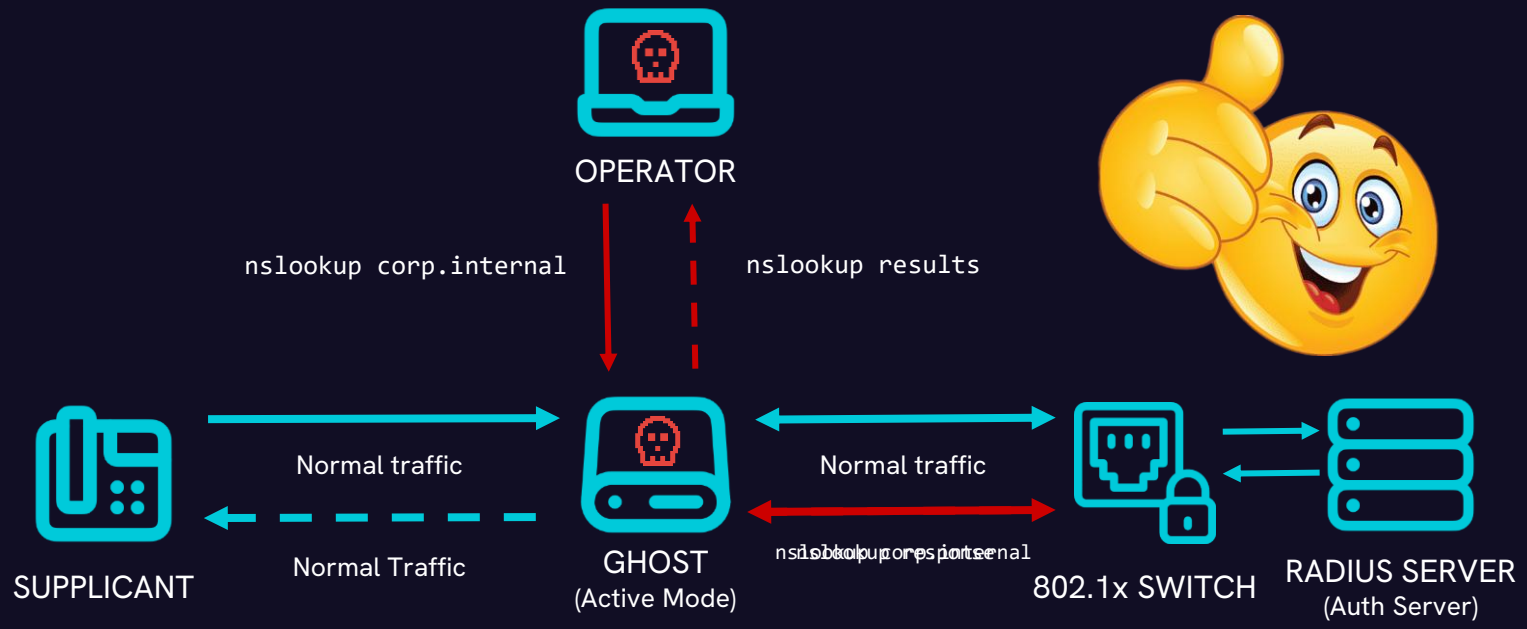
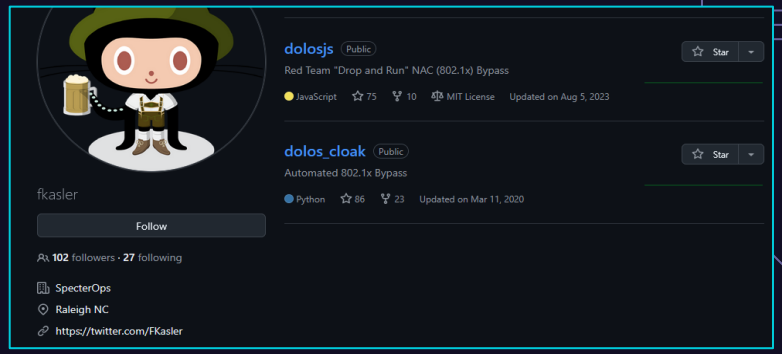


ENHANCING EXISTING TOOLS

03

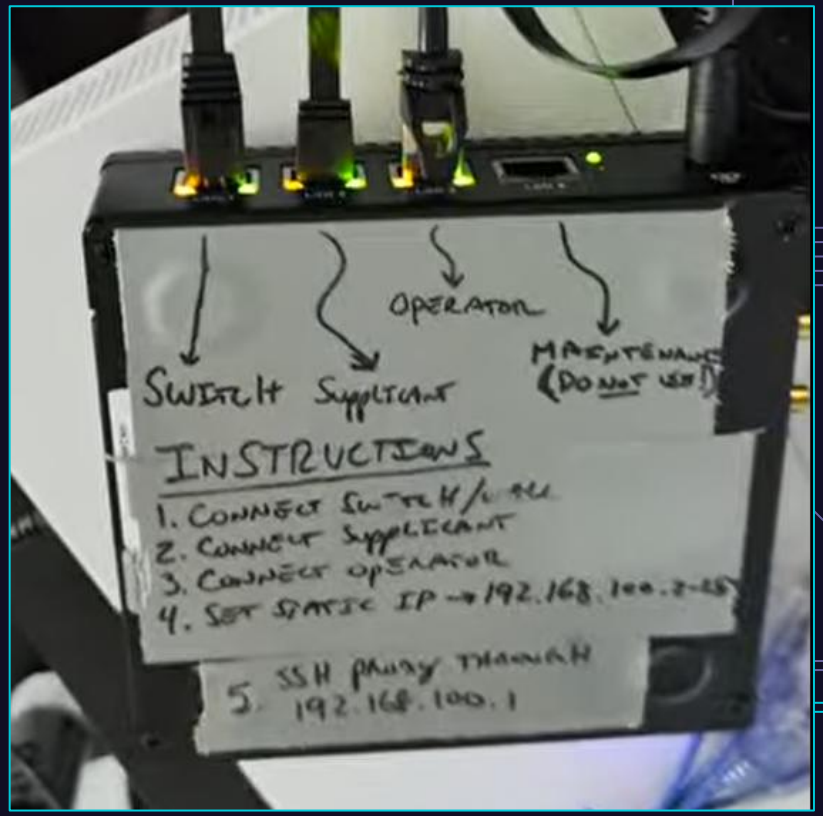
DOLOSJS

- Created by fkasler @ SpecterOps
- Built for NanoPi 2RS w/ Ubuntu Core
- Some outdated elements
- Relay attacks are complex
- Hard-coded install



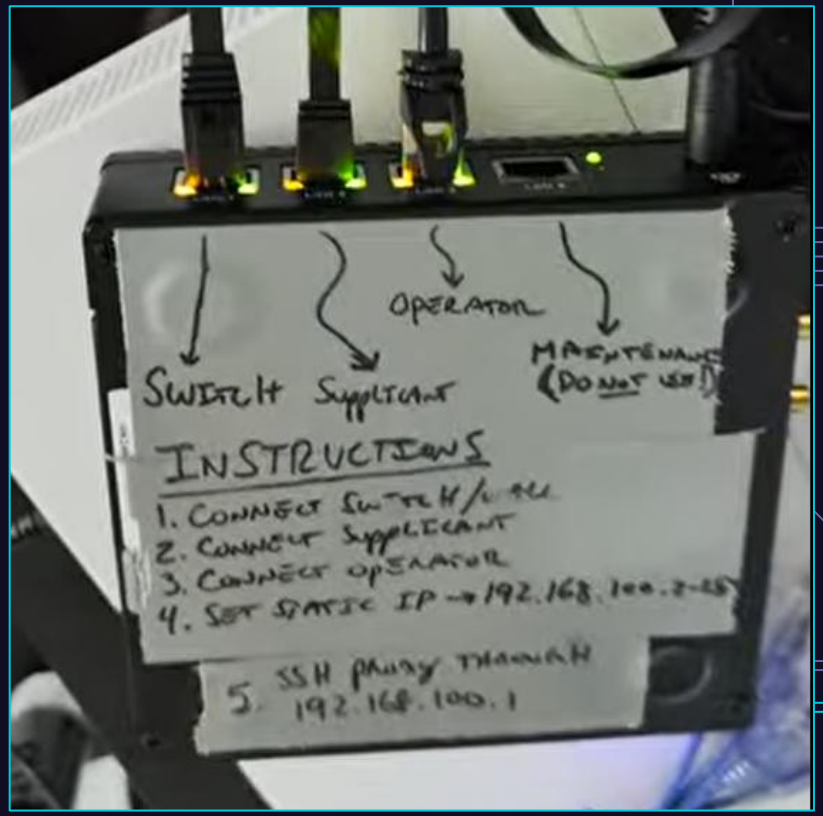
MY IMPLEMENTATION

- Debian 12
- Dynamic installation
- More resources + ports
- Alternate connection method
- “User friendly” instructions
- Looks like a bomb



DRAWBACKS

- Clunky + not portable
- Relay attacks are annoying
- Need to update manually
- **Looks like a bomb**



VERSION 2.0

- Codename: **`Fart Machine`**
- Uses a multi-NIC dongle
- Attack runs directly in attacker VM
- Adapted for netplan

FART MACHINE FEATURES

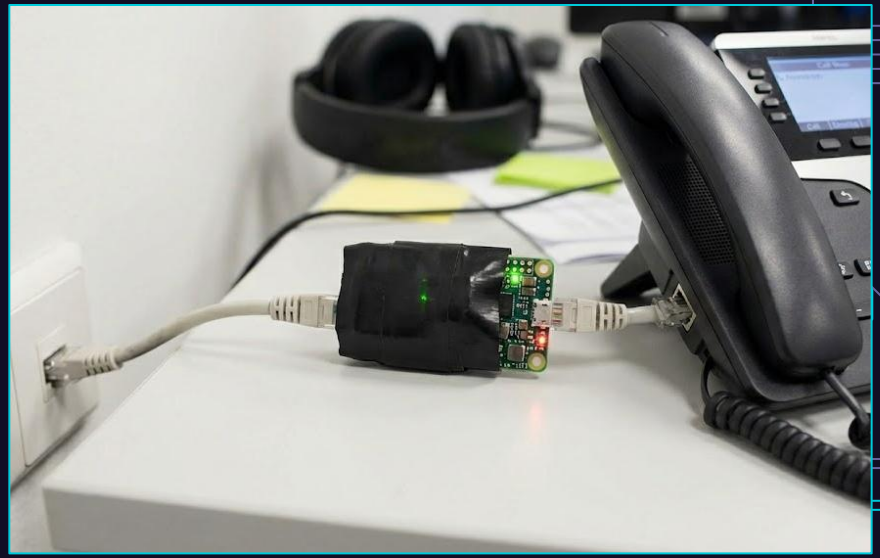
- No external device needed
- No device bottleneck
- Relays are easy now
- Run the attack at will
- Perfect for on-sites

FART MACHINE RESULTS

- Device has consistently bypassed networks with 802.1x NAC
- Bypasses enterprise implementations such as Aruba ClearPass

RED TEAM IMPLANT APPLICATIONS

- Undetectable (digitally)
- No extra port needed



RED TEAM IMPLANT APPLICATIONS

- IP phone with a PoE injector
- AP in a drop ceiling
- Workstation + add badusb

Callback capabilities

- Host an AP
- Connect to C2 over LTE
- LoRa administration/debug

DEFENSE

04

VULNERABILITY TIMELINE

2001

802.1x is Born

The first standard for 802.1x is approved. Developed by 3com, HP, and Microsoft.

2005

Fundamental Flaw Identified

Steve Riley published the first practical hub-based 802.1x bypass.

2010

802.1x-2010 + MACsec

Major revision integrates MACsec + MKA. This prevents traffic forging as each ethernet frame is encrypted/authenticated.

2018

Bypassing MACsec with SilentBridge

Gabriel Ryan developed an attack to bypass MACsec by exploiting a vulnerability in its authentication flow.

2025

The Unfortunate State of NAC

Today, MACsec is still not widely adopted across organizations due to cost and complexity, and it still remains bypassable.

MACsec (802.1AE)

- Media Access Control Security
- Encrypts EVERY frame
- Prevents packet forgery
- It’s like a club with biometrics

WHY IS MACSEC RARE?

- Requires MACsec-capable hardware
- Windows lacks native MACsec support
- IoT devices don’t support MACsec
- Complex for existing networks

WHAT CAN YOU DO?

- Deploy MACsec where possible
- Isolate high-risk devices
- Monitor for traffic anomalies
- Leave no open + active ports

THANK YOU

QUESTIONS?

GitHub

LinkedIn